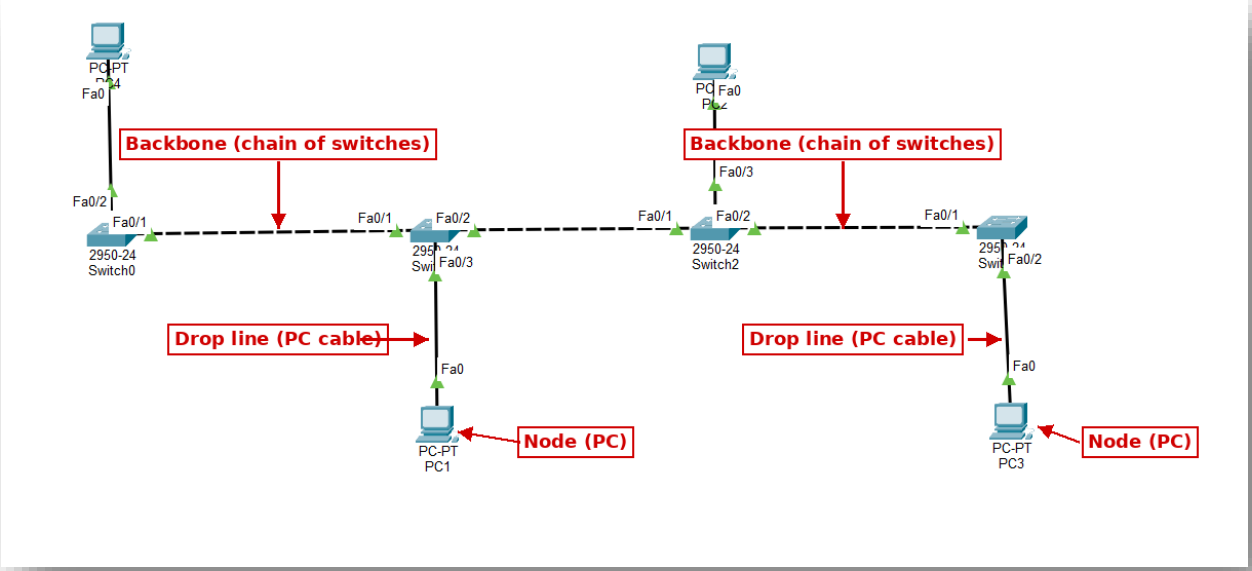
Day 2:-
Network Topologies

Q1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

Answer:



- **Backbone:** the main cable that all devices share.
- **drop lines:** connect each device to the backbone.
- **Nodes:** the computers, server and printer on the network.

In Packet Tracer, the bus topology is shown as a linear chain: the switches joined one after another form the backbone, each PC cable is a drop line, and each PC is a node.

Real-world example: A small computer lab in an old Ethernet network (10BASE2), where all computers are connected to one shared coaxial cable. Another example is the CAN bus inside a car, where many control units share one common line.

Q2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

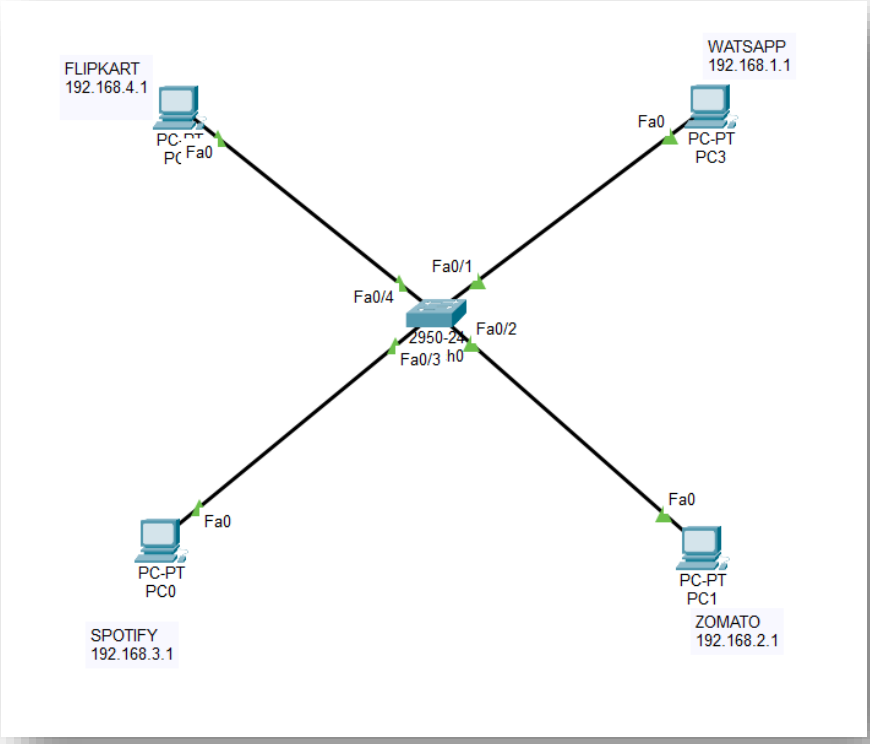
Answer:

Steps in Cisco Packet Tracer:

1. Open Packet Tracer. In the bottom-left device box choose Network Devices > Switches and drag a 2950-24 switch to the workspace.
2. Choose End Devices and drag 4 PCs to the workspace.
3. Choose Connections > Copper Straight-Through cable. Click a PC (FastEthernet0), then click the switch (FastEthernet0/1). Repeat for the other PCs using Fa0/2, Fa0/3 and Fa0/4.
4. Rename each PC: click the PC, open the Config tab and type the name in Display Name (WhatsApp, Zomato, Spotify, Flipkart).
5. Give each PC an IP address: click the PC > Desktop > IP Configuration > Static, and enter the values from the table below.
6. Wait until all link lights turn green, then test with Desktop > Command Prompt > ping 192.168.1.2.

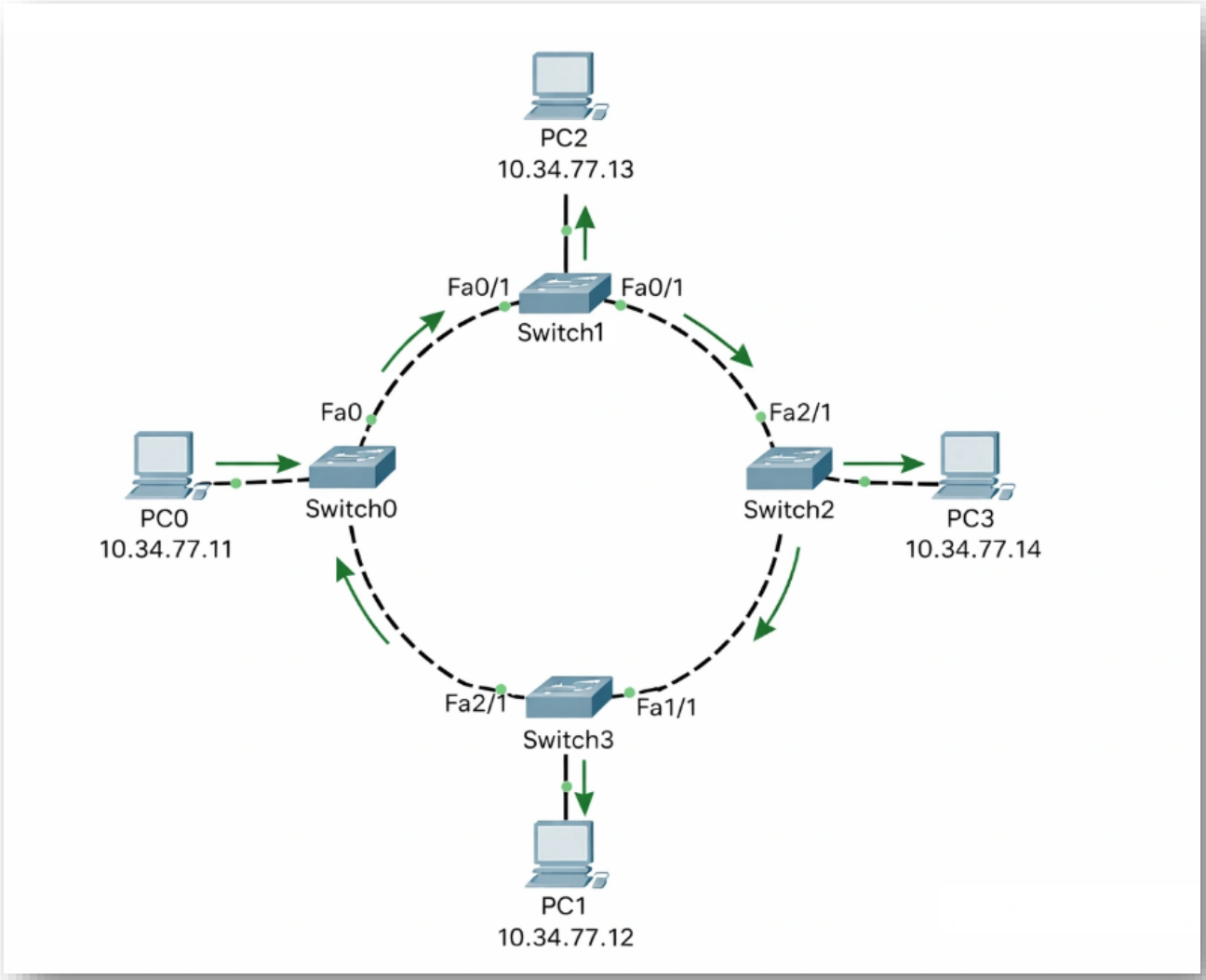
PC name	IP address	Subnet mask	Connected to
WhatsApp	192.168.1.1	255.255.255.0	Switch Fa0/1
Zomato	192.168.1.2	255.255.255.0	Switch Fa0/2
Spotify	192.168.1.3	255.255.255.0	Switch Fa0/3
Flipkart	192.168.1.4	255.255.255.0	Switch Fa0/4

Topology plan:



Q3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Answer:



Explanation: In a ring topology each device is connected to exactly two neighbours, and data moves in one direction around the ring. When PC A sends data to PC D, the data goes first to PC B, which checks the address and passes it on, then to PC C, and finally to PC D, which accepts it. Many rings use a token, so a device can send only when it holds the token, which avoids collisions.

Q4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Answer:

Topology	Advantages	Disadvantages
Star	1. Easy to add or remove devices, and easy to find faults. 2. If one cable or device fails, the rest of the network keeps working.	1. If the central switch fails, the whole network stops. 2. Needs more cable and a switch, so the cost is higher than a bus.
Mesh	1. Very reliable, because there are many alternate paths if a link fails. 2. Fast, because every device has direct links and there is no traffic bottleneck.	1. Very expensive, because it needs many cables and ports (5 devices need 10 links). 2. Complex to set up and manage as devices increase.